VOL. IV — ESTABLISHED MMXXIV

The SNS Conference Tool

A discreet register connecting researchers at conferences through curated affinities of inquiry.

USER MANUAL
FOR FELLOWS & THE REGISTRY · MMXXVI

Contents

Section One

Quick start

A two-minute introduction to the SNS Conference Tool — for the fellow who would rather try it than read about it.

YOU WANT TO…	DO THIS
Try the app right now	Open `http://localhost:3000` → **A Returning Fellow** → log in `you@example.com` / `Demo!2026`
Browse the conference floor	Tap **Discover** → tap **Join** on NeurIPS 2026 Bangkok
See who is nearby	Tap **Fellows** in the bottom bar → look at Intellectual Affinities
Send a message	Tap a fellow card → **Begin correspondence**
Manage events / users	(Admin only) tap **Registry** in the bottom bar

> **DEMO DATASET**
>
> The app ships with twenty seeded fellows, three sessions, ~163 affinities, and nineteen chat messages — everything in this manual is reproducible without a real conference. See §7 for the seeded login choices.

Section Two

Getting in

The app's front door is a single URL. From there, two paths lead to the rest of the system.

THE WELCOME SCREEN

Both pathways begin at the welcome page. **Request Admission** opens the registration funnel; **A returning fellow** goes straight to login.

The page sets the visual tone for everything that follows: serif headlines, brass capitals, a discreet hairline rule. This same chrome carries through every screen of the app.

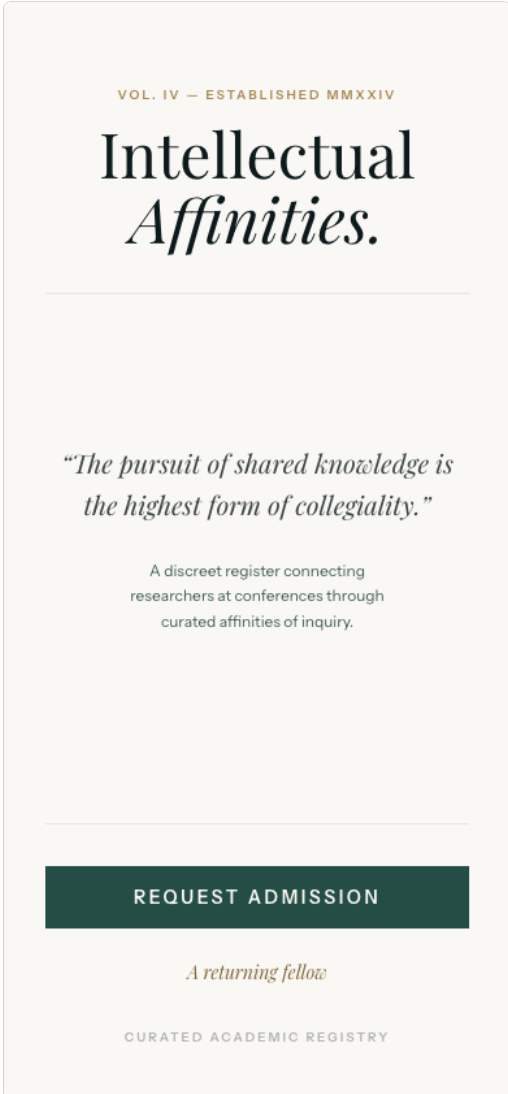

Figure 1 · Welcome

2.1 First-time admission

Tap **Request Admission** on the welcome screen, then enter your **work email**. We send a six-digit transaction number (TAN) to that address — in dev mode the TAN is always `123456`.

After verifying the TAN, fill in your **particulars**: first name, last name, academic title (Prof. / Dr. / PhD candidate / etc.), institution, and a password (≥ 8 characters, must not equal your email's local part, and must not be one of the embedded common-password blocklist).

Tap **Confirm Enrolment**. You're now logged in and dropped on the **Discover** screen.

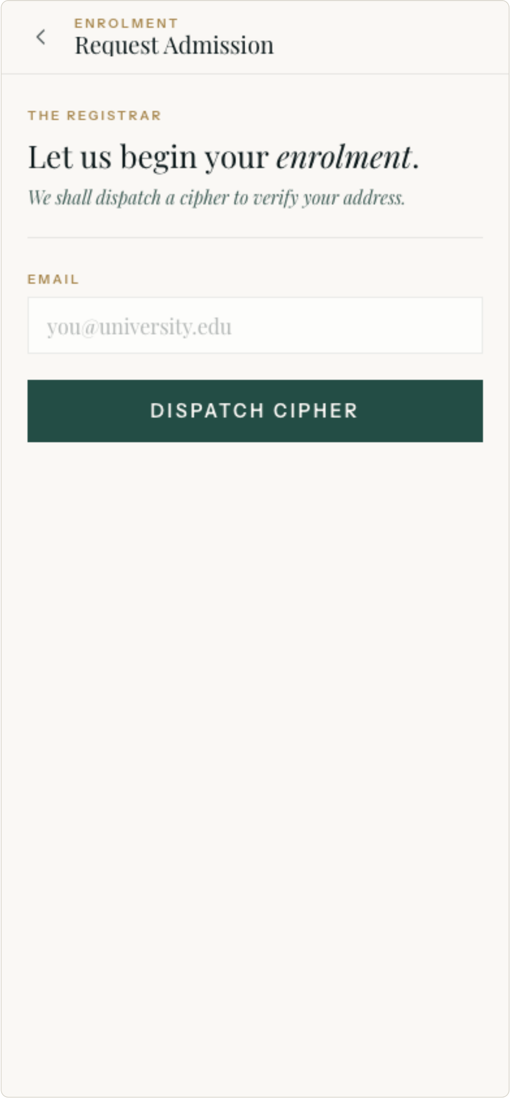

Figure 2 · Request Admission

2.2 Returning fellow

If you already have an account, tap **A returning fellow** on the welcome screen, enter your email + password, and tap **Grant Admission**.

Wrong password, unknown email, or a suspended account all return the same generic "Admission refused" message — this is intentional, so attackers cannot enumerate registered accounts.

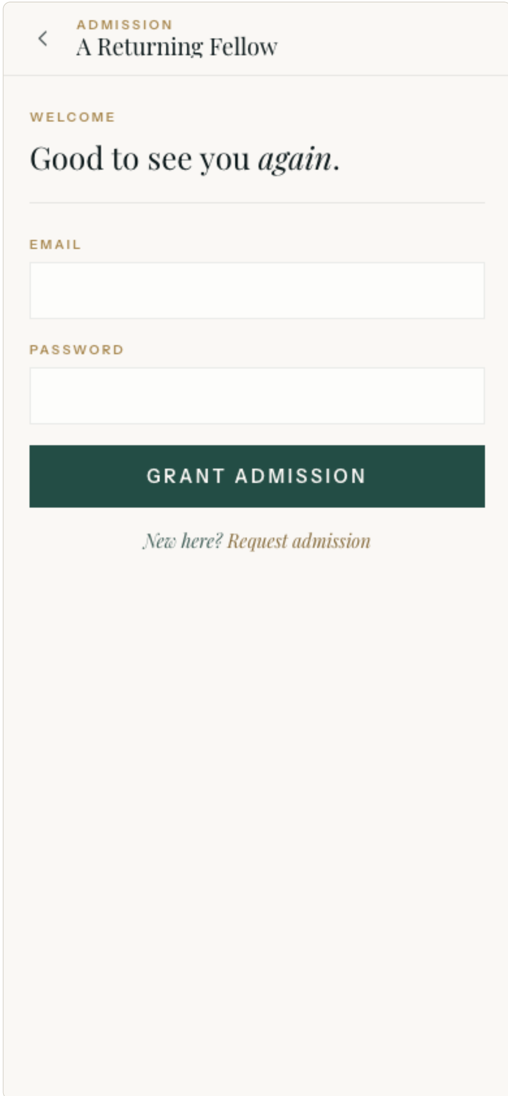

Figure 3 · Login

FORGOT YOUR PASSWORD?

v1 of the system does not ship a self-service password reset. Contact your event organiser; they can suspend / reinstate your account from the **REGISTRY** (see §6.4).

Section Three

The participant app

After login you land on the **Discover** screen — the conference home. Everything below describes the screens you can reach from there.

3.1 The bottom tab bar

A persistent tab bar lives at the bottom of every authenticated screen. Four tabs by default; admins see a fifth, **Registry**.

TAB	WHERE IT GOES	WHAT LIVES THERE
Discover	`/events/join`	Join a session by QR or code.
Fellows	`/matches`	Your historical affinities (and the live vicinity per session).
Letters	`/chats`	All your active chat threads.
Study	`/settings`	Settings, profile, interests, sign-out.
Registry (admins only)	`/admin`	The management console (see §6).

3.2 Discover — joining a session

A "session" is a conference event the organiser has set up (e.g. NeurIPS 2026 Bangkok).

You can join by **scanning the QR cipher** printed on your event badge, or by typing the cipher into the Or transcribe field, or by tapping **Join** on a known session in the Demo Sessions list below the scanner.

On join, you are added to the session roster with a default **50 m** visibility radius, and your GPS is requested (you may decline; vicinity will not work for that session). Once joined, the session becomes your active session — tapping **Fellows** now shows live nearby fellows.

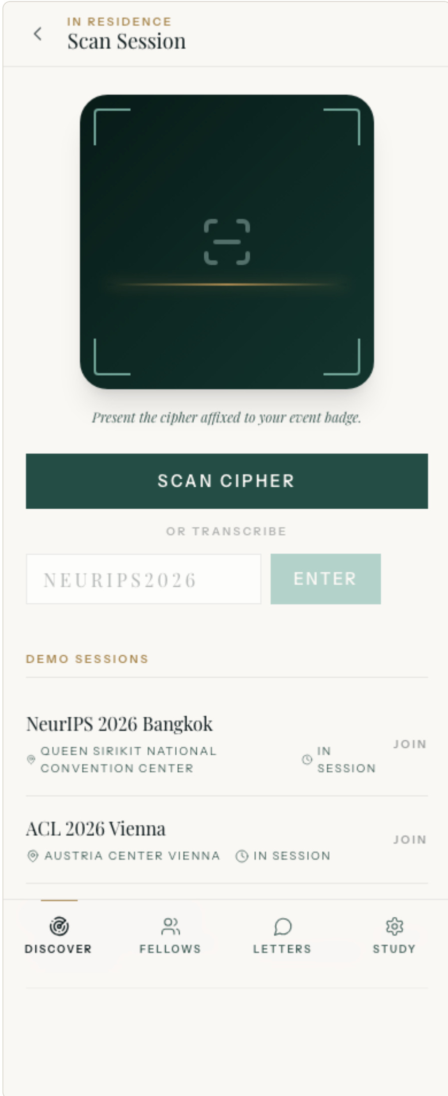

Figure 4 · Discover

3.3 Fellows — your match register

The **Fellows** tab is the heart of the app. It lists everyone in your active session ranked by intellectual affinity (cosine similarity of interest profiles) and, secondarily, by physical proximity.

Use the **radius selector** (20 m / 50 m / 100 m) to control how far the vicinity query reaches. Use the **By Affinity** dropdown to switch between sort orders (Affinity / Proximity / Recency). Use the **Mutual / All** chips to hide one-way matches.

Each card shows a portrait, academic title, name, institution, the keywords you share with that fellow, the distance between you, and an **INDEX** percentage representing the cosine score.

Tap any card to open the Match Dossier with the full keyword overlap and a **Begin correspondence** button.

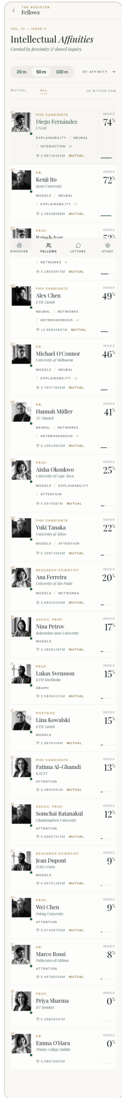

Figure 5 · Fellows (live vicinity)

3.4 Letters — chat correspondence

The **Letters** tab lists every chat thread you have, most recent first. Each row shows the other fellow's portrait, name, institution, the last message preview (prefixed with You: if you sent it), the time delta, and an unread badge when you have messages addressed to you that you haven't read yet.

Open a thread to see the full message history. Type your reply in the bottom composer and press Enter. Messages are delivered in real time over a STOMP websocket — the other party sees them appear without refreshing.

If you have granted notification permission on a mobile device, you'll get an OS notification when a new message arrives while the app is in the background.

Figure 6 · Letters

3.5 Study — settings, profile, interests

Tap **Study** in the bottom tab to reach the settings page.

- **Profile** — edit name, academic title, institution, profile portrait.
- **Interests** — add / remove the interests that drive matching (see §4).
- **SNS Links** — optionally connect Facebook or LinkedIn so the system can enrich your profile automatically.
- **Push notifications**, **GPS consent**, **Local storage**, **Language** — toggle on / off.
- **Sign out** — clears your session.
- **Export my data** — downloads a ZIP of every record we have about you (§5).
- **Delete my account** — schedules permanent deletion in 30 days (§5).

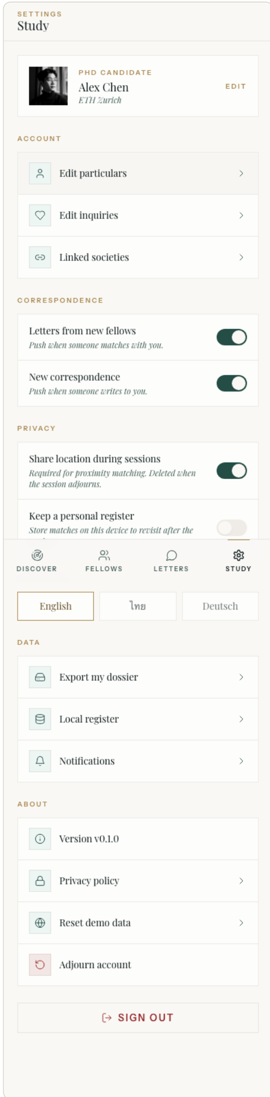

Figure 7 · Study (settings)

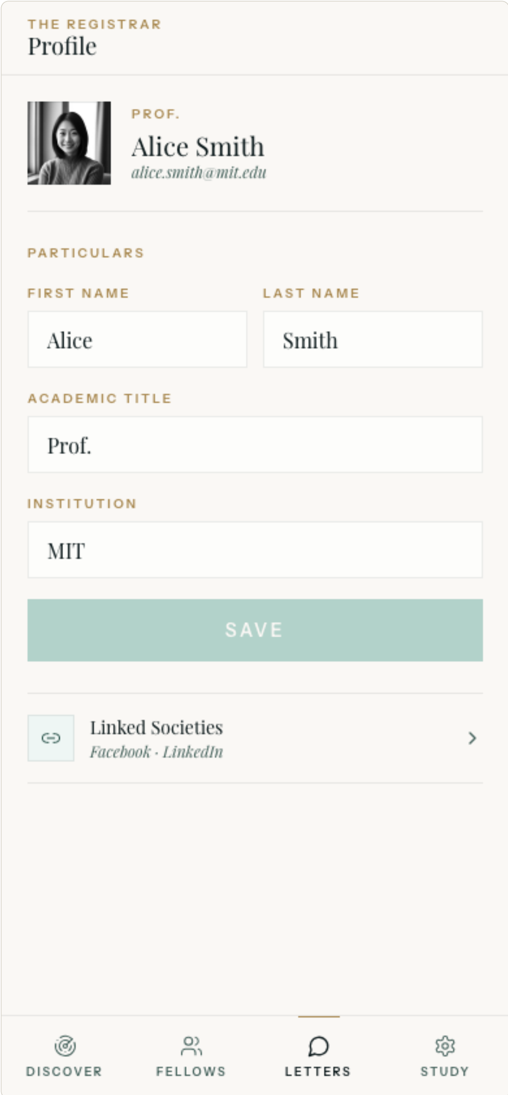

Figure 8 · Profile

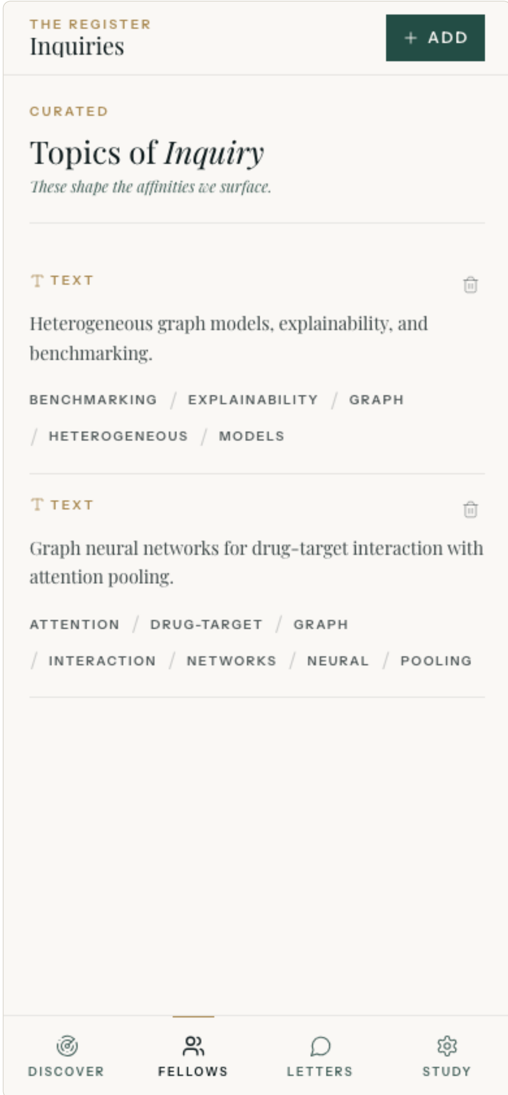

Figure 9 · Interests

Section Four

Interests & matching

Matching is a cosine similarity between your keyword vector and every other fellow's, scoped to the session you're in. To produce useful matches, the system needs to know what you work on.

4.1 Three input modes

1. **Free text.** Type a sentence or paragraph (e.g. "I work on graph neural networks for drug-target interaction prediction, focusing on heterogeneous graphs and explainability."). The keyword extractor pulls out the concept-bearing terms and weights them.
2. **Article link.** Paste a URL (e.g. an arXiv abstract). The system stores the link and the descriptive text you provide alongside it.
3. **Article upload.** Upload a PDF or plain-text article. The text is extracted and run through the same keyword pipeline.

4.2 Practical advice

- **Quality over quantity.** Two or three rich, specific paragraphs produce better matches than a long list of single keywords.
- **Use natural language.** "Federated learning, privacy, distributed training" works, but "Federated learning for medical imaging with strict privacy guarantees over hospital networks" is much better — the extractor uses bigram phrases too.
- **Edit anytime.** Removing or adding interests triggers an automatic re-computation of your matches for every session you've joined.

> **INSPECT YOUR KEYWORDS**
>
> Open the
> **INTERESTS**
> screen — each interest card lists the keywords it contributed to your vector. Useful for debugging when you wonder why two researchers don't appear as similar as you'd expect.

Section Five

Privacy & GDPR

The system is built for compliance from the database upward — your data is yours.

5.1 What we store, and how

- **Locations** are only stored as long as a session is active and only at coarse precision (~1 m).
- **IP addresses** are SHA-256-hashed with a per-deployment salt before being written to the audit log — raw IPs never persist.
- The **audit log** is append-only, enforced by a Postgres trigger, and pruned automatically after 180 days.
- **Encrypted SNS tokens** are AES-256-GCM at rest; if you unlink the account, the tokens are zeroed.

5.2 Your rights

ACTION	WHERE	EFFECT
Export everything	Study → Export my data	Streams a ZIP: `profile.json`, `interests.json`, `matches.json`, `chat-threads.json`, `chat-messages.json`, `sns-links.json`, `manifest.json`.
Soft delete	Study → Delete my account	Sets `deleted_at = now()`. You're immediately logged out and can no longer log in.
Hard delete	(automatic, after 30 days)	A scheduled cron permanently removes every row tied to your `user_id`; cascades wipe profile, interests, participations, matches, chat, devices, refresh tokens, SNS links.

You can also ask an admin to suspend or hard-delete you immediately — see §6.4.

Section Six

The Registry — admin

For event organisers, support staff, and operations. Only users with role `ADMIN` or `SUPER_ADMIN` can reach the management console.

THE SAME CHROME, THE SAME SHELL

The console **shares the participant app's chrome** — same cream theme, same top app bar, same bottom tab bar. The only structural difference is a horizontal section nav of pills below each page header for jumping between admin sub-sections.

The bottom tab bar grows from four tabs to five when your role is `ADMIN` or `SUPER_ADMIN`; the new **Registry** tab is the canonical entry. Admins still land on the participant home (`/events/join`) on login like everyone else.

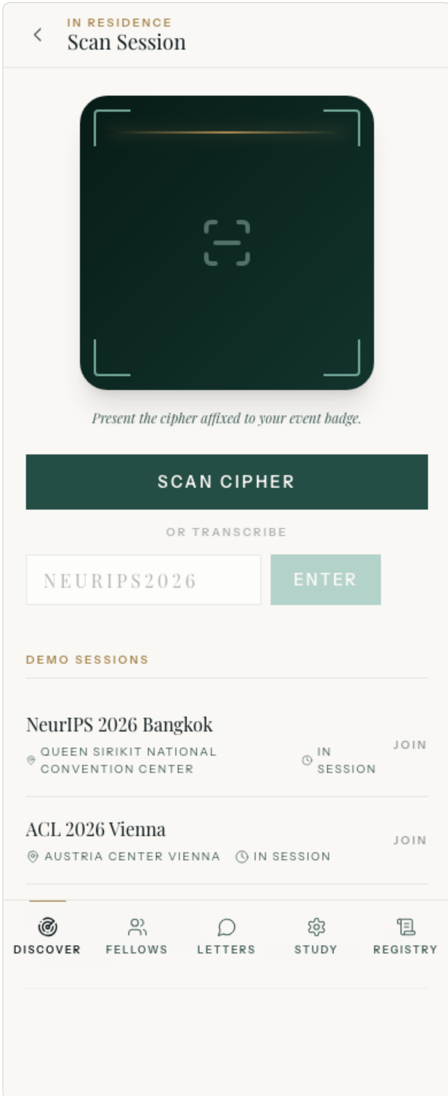

Figure 10 · Discover, with the Registry tab visible

6.1 Becoming an admin

1. **You are the seeded demo super-admin.** `you@example.com` ships as `SUPER_ADMIN` on a fresh seed; `lukas.svensson@kth.se` and `rajesh.iyer@stanford.edu` ship as `ADMIN`.

2. **An operator promoted you.** Any existing `SUPER_ADMIN` can change your role on the Fellows admin screen (§6.4).
3. **You are the bootstrap admin.** Setting the env var `SNS_ADMIN_EMAIL=you@whatever.com` and rebooting the backend automatically promotes that account to `SUPER_ADMIN` on every boot. (In production, `ProductionSecretsCheck` refuses to start without it.)

6.2 Overview

The first stop in the Registry. A tile-grid dashboard, refreshes every 30 seconds.

- **Fellows**: total · active · suspended · deleted (24 h)
- **Sessions**: active · expired · affinities · new (24 h)
- **Apparatus**: outbox pending · outbox failed · delivered (24 h) · audit (24 h)

Tiles turn **amber** when the value enters a warning band (e.g. > 50 outbox pending) and **red** when there's an actual failure (e.g. any outbox FAILED).

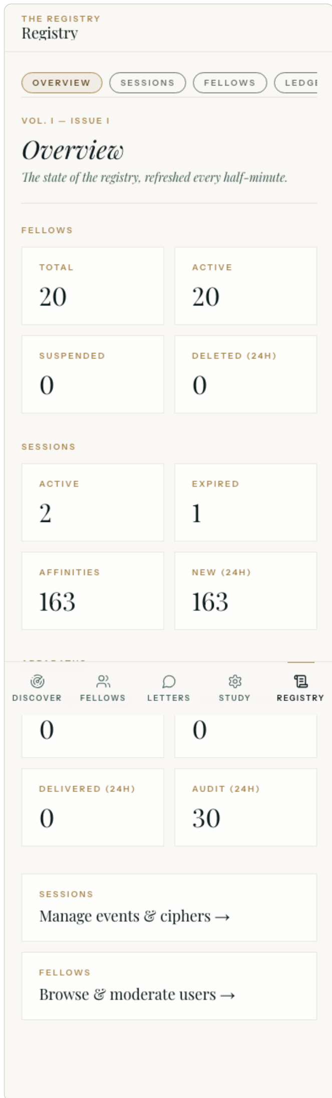

Figure 11 · Registry · Overview

6.3 Sessions

The conference-organiser surface. The list view is a paged table of every session — name, venue, cipher, status, and fellow count — with a search box across name / venue / cipher and a **+ New** button to create.

Tap a row for the session detail.

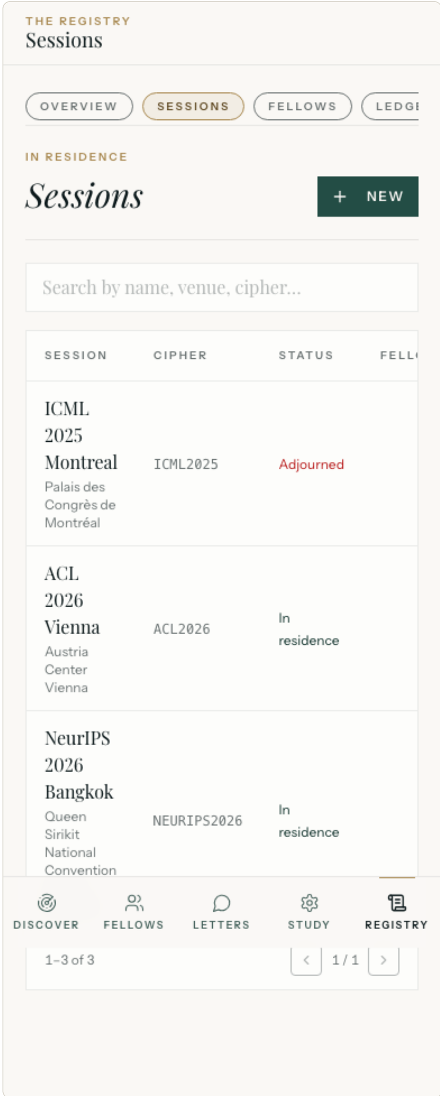

Figure 12 · Sessions

CREATE NEW SESSION

Use the **+ New** button on the Sessions list. Enter a **name**, **venue**, **cipher** (the QR plaintext, e.g. `NEURIPS2027`), and an **adjournment time**. Optionally provide a **centroid lat/lon** to centre the heatmap on a fixed point.

Tapping **Create session** redirects you to the session detail with the new cipher ready to print on badges.

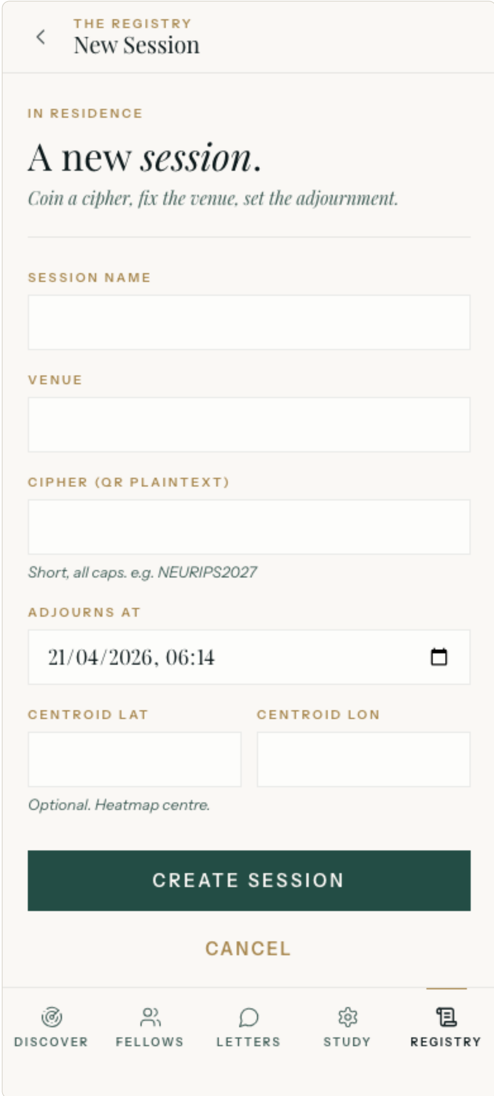

Figure 13 · New session

SESSION DETAIL

Header shows name + venue + cipher. Stat tiles count fellows in residence, affinities, correspondence (chat messages), and overall status.

Below the stat tiles is a **Cipher QR card** — the cipher plaintext sits above a scannable QR code (Editorial Ivory ink on cream paper). Two outline buttons download the QR for badge printing: **PNG** exports `<CIPHER>-qr.png` at 1024×1024 px on white background (≈ 8.7 cm at 300 dpi); **SVG** exports `<CIPHER>-qr.svg` for crisp scaling at any size. Both encode the plaintext cipher; participants can scan or type. Error-correction level H tolerates smudges or partial coverage on the printed badge.

The **venue heatmap** is an inline SVG with concentric 10/25/50 m rings centred on the venue, with each fellow's last known position dotted. Useful for spotting clusters or stragglers in real time.

Below the heatmap is the **Fellows present** table — name + institution + last GPS fix time + selected radius. Tap a row to jump to that fellow's dossier.

The **Adjourn permanently** button at the bottom hard-deletes the event after a confirmation tap; the deletion cascades to participations, matches, and chat messages. It cannot be undone.

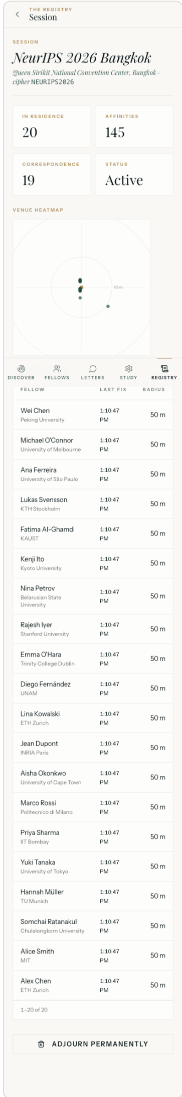

Figure 14 · Session detail

6.4 Fellows

The user-admin surface. Paged list of every user with email + name + institution + role + status, plus a search box and two filter selectors: by role (USER / ORGANIZER / ADMIN / SUPER_ADMIN) and status (active / suspended / deleted).

Tap a row to open the dossier.

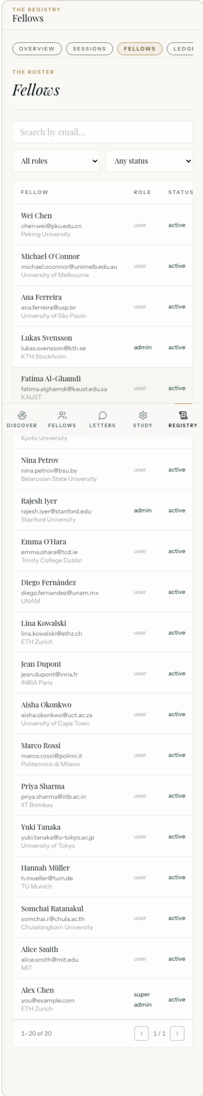

Figure 15 · Fellows

FELLOW DOSSIER

Header shows portrait, full name, email, academic title, institution. Six stat tiles count: status, role, interests, affinities, messages, sessions.

Role chips let you change role with a single tap; greyed if it's already the current role. SUPER_ADMIN-only; refused if it would demote the last super-admin.

Below: lists of **interests** (with extracted keywords), **sessions joined** (with selected radius and join date), and the latest 50 **ledger entries** where this fellow was the actor.

Action buttons at the bottom: **Suspend / Reinstate**, **Soft delete**, **Hard delete**. Hard delete is two-tap (first tap shows Confirm hard delete) and SUPER_ADMIN-only.

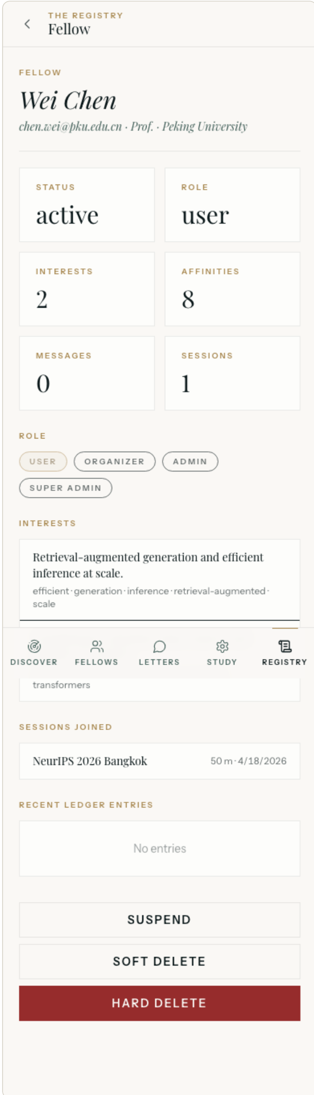

Figure 16 · Fellow dossier

6.5 Ledger

Read-only search over the immutable `audit_log` table (a Postgres trigger blocks UPDATE / DELETE outside the prune job).

Three optional filters: an **actor user id** (paste a UUID from a fellow dossier), an **action code** (e.g. `auth.login.failure`, `admin.user.suspended`), and a **since** datetime lower bound. The list pages 50 rows at a time.

The 24-hour total appears as a tile on the Overview and Apparatus screens.

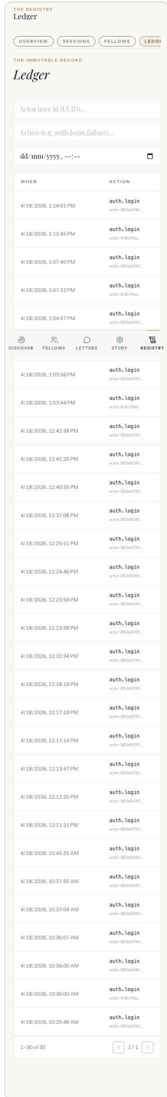

Figure 17 · Ledger

COMMON AUDIT ACTIONS

CODE
`auth.register` / `auth.verify` / `auth.complete`
`auth.login` / `auth.login.failure` / `auth.login.suspended` / `auth.login.unverified`
`auth.refresh` / `auth.refresh.reuse_detected`

CODE
`auth.logout`
`profile.update` / `profile.soft_delete` / `profile.hard_delete`
`sns.link` / `sns.callback` / `sns.unlink` / `sns.enrich`
`export.download`
`admin.event.created` / `admin.event.updated` / `admin.event.deleted`
`admin.user.suspended` / `admin.user.unsuspended` / `admin.user.role_changed` / `admin.user.soft_deleted` / `a
`admin.outbox.retry`

6.6 Apparatus

The push pipeline + system-health surface. Top tiles repeat the Apparatus row from Overview: **outbox pending**, **outbox failed**, **delivered (24 h)**, and **audit (24 h)**.

Filter chips above the table: **all · PENDING · FAILED · DELIVERED**. The table shows kind + status + attempt count, with a **Retry** link on FAILED rows that flips them back to PENDING for the next scheduled drain.

The refresh icon in the header forces an immediate reload of both metrics and the outbox table.

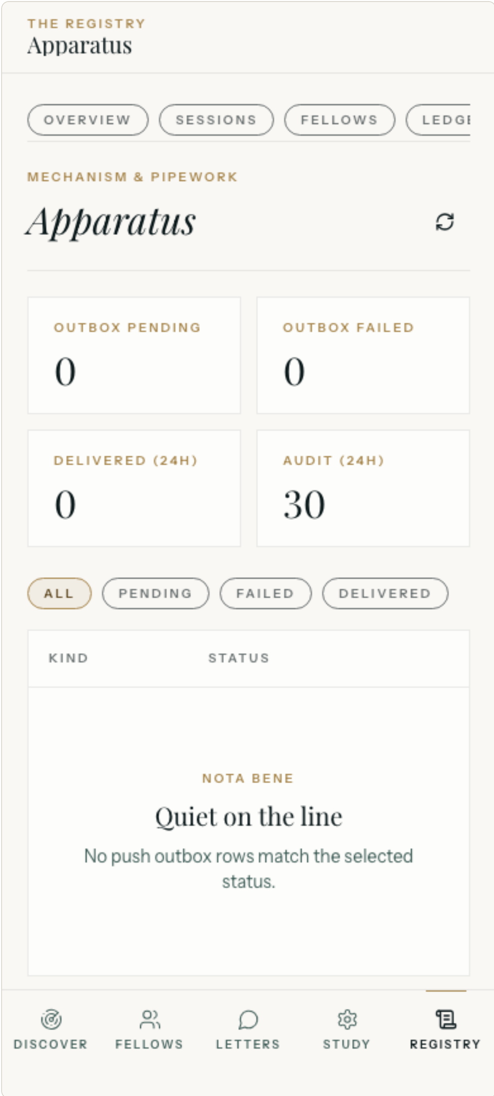

Figure 18 · Apparatus

Section Seven

Demo accounts

The seeded development dataset gives you twenty ready-made fellows and three sessions to experiment with. Every account uses the same password.

EMAIL	ROLE	NOTES
`you@example.com`	SUPER_ADMIN	Alex Chen, ETH Zurich. Already joined NeurIPS + ACL with interests + chat history.
`lukas.svensson@kth.se`	ADMIN	Lukas Svensson, KTH Stockholm.
`rajesh.iyer@stanford.edu`	ADMIN	Rajesh Iyer, Stanford University.
`alice.smith@mit.edu`	USER	Alice Smith, MIT. Use this to test what a regular fellow sees.
(any of the other 16 seeded fellows)	USER	All seeded with the same password.

Password for all accounts: `Demo!2026`

> **RE-SEEDING**
> The seeder runs once on a fresh DB volume and short-circuits on subsequent boots when `you@example.com` already exists. To re-seed, wipe the volume:
>
> ```
> docker compose -f infra/docker-compose.dev.yml --profile backend --profile web down -v
> docker compose -f infra/docker-compose.dev.yml --profile backend --profile web up -d --build
> ```

Section Eight

Troubleshooting

"ADMISSION REFUSED" WITH THE RIGHT PASSWORD

Almost always one of:

1. **The backend was restarted** and it rotated its ephemeral JWT keypair (dev only — prod uses persistent keys). The frontend tries to refresh with the stale token and gets 400. **Fix:** clear browser localStorage (DevTools → Application → Storage → Clear site data) and log in again.
2. **You hit the rate limit** — login allows 30 attempts per IP per hour and 10 per email per hour. **Fix:** wait, or restart the backend (`docker restart sns-backend`) to drop the in-memory limiter.
3. **Your account was suspended.** A super-admin needs to reinstate you in Registry → Fellows.

FELLOWS PAGE IS BLANK AFTER JOINING

- **Wait 30 seconds.** Matching recomputes asynchronously after you join; vicinity refreshes every 30 seconds.
- **Widen the radius** to 100 m.
- **Add interests.** Without interests you have no keyword vector — no peer can have a non-zero similarity with you.
- **Check your GPS.** Without a recent location fix (within 5 minutes) the vicinity query won't include you. Check Study → GPS consent is on.

CHAT MESSAGES DON'T ARRIVE IN REAL TIME

- The websocket auth needs your JWT — sign out and back in if your session is stale.
- Page reload often kicks the STOMP client back into a working state.
- The Letters thread list refreshes every 15 seconds even without WS, so you'll see new messages on a refresh.

Section Nine

Glossary

TERM	MEANING
Admission	The login screen ("A Returning Fellow").
Adjourn	Either let a session expire (passive) or delete a session (admin action).
Affinity	The cosine-similarity score between two fellows' interest vectors.
Apparatus	The admin ops dashboard (push outbox + system health).
Cipher	The plaintext QR code that joins a session (e.g. `NEURIPS2026`).
Correspondence	A chat thread between two fellows in the same session.
Discover	The participant home screen — list of joinable sessions.
Dossier	The detailed view of a single fellow on the admin Fellows screen.
Fellow	Any registered user.
Index	A fellow's affinity score expressed as a 0–100% percentage.
In residence	The session's status while it's active (not adjourned).
Ledger	The admin audit-log explorer.
Letters	The participant chat inbox tab.
Particulars	Your profile data (name, title, institution, portrait).
Registry	The admin console as a whole; also the bottom tab that opens it.
Session	A single conference event (NeurIPS Bangkok, ACL Vienna, ...).
Study	The participant settings tab.
TAN	The six-digit transaction number used to verify your email at registration.
Vicinity	The set of fellows currently within your selected radius (20 / 50 / 100 m).

End of manual · Vol. IV, Issue I · MMXXVI